

P1. (a)

$$V^\pi(s) = \sum_{a \in A} \pi(a|s) Q^\pi(s, a) \text{ --- by total probability theorem.}$$

then we consider the best action $a_{\max}$ s.t. $Q^*(s, a_{\max}) = \max_{a \in A} Q^*(s, a)$.

Thus, $\forall \pi$,

$$V^\pi(s) = \sum_{a \in A} \pi(a|s) Q^\pi(s, a) \leq \sum_{a \in A} \pi(a|s) Q^*(s, a_{\max}) = 1 \cdot Q^*(s, a_{\max}) = \max_{a \in A} Q^*(s, a)$$

The inequality can also be apply to $V^*(s)$,

now we know $V^*(s) \leq \max_a Q^*(s, a)$, then we consider the inequality $V^\pi(s) \leq \max_a Q^*(s, a)$,

by definition of $V^*(s)$, if $V^*(s) < \max_a Q^*(s, a)$, then the previous state of s (call it s')

will have a greater value function, call it $V^\phi(s) = \max_a Q^*(s, a) > V^*(s)$,

but if such ϕ exist, $V^*(s)$ cannot be the maximum over all possible policy, Hence, $V^*(s) = \max_a Q^*(s, a)$

$$(2) Q^*(s, a) = \max_{\pi \in \Pi} Q^\pi(s, a)$$

$$= \max_{\pi \in \Pi} (R_{s,a} + \gamma \sum_{s' \in S} P_{ss'}^a V^\pi(s'))$$

$$= R_{s,a} + \gamma \sum_{s' \in S} P_{ss'}^a \max_{\pi \in \Pi} V^\pi(s')$$

$$= R_{s,a} + \gamma \sum_{s' \in S} P_{ss'}^a V^*(s').$$

(b)

(∞ -norm)

$$\|T^*(Q) - T^*(Q')\|_\infty = \max_{(s,a)} |T^*(Q)(s,a) - T^*(Q')(s,a)|$$

$$= \max_{(s,a)} |R_{s,a} + \sum_{s'} \gamma P_{ss'}^a \max_{a'} Q(s', a') - R_{s,a} - \sum_{s'} \gamma P_{ss'}^a \max_{a'} Q'(s', a')|$$

$$= \max_{(s,a)} |\gamma \sum_{s'} P_{ss'}^a (\max_{a'} Q(s', a') - \max_{a'} Q'(s', a'))| \leq \max_{(s,a)} \max_a |\gamma \sum_{s'} P_{ss'}^a (Q(s', a') - Q'(s', a'))|$$

$$= \max_{(s,a)} \max_a |\gamma (Q(s', a') - Q'(s', a'))| \leq \gamma \|Q - Q'\|_\infty$$

(l_1 -norm)

$$\|T^*(Q) - T^*(Q')\|_1 = \sum_{(s,a)} |T^*(Q)(s,a) - T^*(Q')(s,a)|$$

$$\leq |S||A| \max_{(s,a)} |T^*(Q)(s,a) - T^*(Q')(s,a)|$$

$$= |S||A| \gamma \|Q - Q'\|_\infty$$

$$\leq \gamma |S||A| \gamma \|Q - Q'\|_1, \text{ Here our contraction operator is } \gamma |S||A|,$$

which is usually greater than 1, so the property does not hold for l_1 -norm.

P2.

$$\begin{aligned} \frac{1}{1-\gamma} E_{s \sim d_{\mu}^{\pi_0}} E_{a \sim \pi_0(\cdot|s)} [f(s,a)] &= \frac{1}{1-\gamma} \sum_{s \in S} d_{\mu}^{\pi_0}(s) \sum_{a \in A} \pi_0(a|s) \cdot f(s,a) \\ &= E_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t p(s_t = s | s_0, \pi_0) \sum_{a \in A} \pi_0(a|s) \cdot f(s,a) \right] \\ &= \sum_{t=0}^{\infty} \gamma^t p_{\pi_0}^{\mu}(t) \cdot f(s,a) = E_{T \sim p_{\pi_0}^{\mu}} \left[\sum_{t=0}^{\infty} \gamma^t f(s,a) \right]. \end{aligned}$$

P3.

$$\pi_0(a|s) = \frac{1}{1+5+4} = \frac{1}{10}, \quad \pi_0(b|s) = \frac{5}{1+5+4} = \frac{5}{10}, \quad \pi_0(c|s) = \frac{4}{1+5+4} = \frac{4}{10}.$$

(a.)

$$\begin{aligned} E[\hat{\nabla} V] &= E_{\tau \sim p_{\pi_0}^{\mu}} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_0}(s_t, a_t) \nabla_{\theta} \log \pi_0(a_t | s_t) \right] \\ &= 0.1 \nabla_{\theta} \log \pi_0(a|s) + 0.5 \nabla_{\theta} \log \pi_0(b|s) + 0.4 \nabla_{\theta} \log \pi_0(c|s) \\ &= 0.1 \cdot 100 \begin{bmatrix} -0.1 \\ -0.5 \\ -0.4 \end{bmatrix} + 0.5 \cdot 98 \begin{bmatrix} -0.1 \\ 1-0.5 \\ -0.4 \end{bmatrix} + 0.4 \cdot 95 \begin{bmatrix} -0.1 \\ -0.5 \\ 1-0.4 \end{bmatrix} \\ &= \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{Var}[\hat{\nabla} V] &= (0.1) \left(100 \begin{bmatrix} 0.9 \\ -0.5 \\ -0.4 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right) \left(100 \begin{bmatrix} 0.9 \\ -0.5 \\ -0.4 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right)^T \\ &\quad + (0.5) \left(98 \begin{bmatrix} -0.1 \\ 0.5 \\ 0.4 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right) \left(98 \begin{bmatrix} -0.1 \\ 0.5 \\ 0.4 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right)^T \\ &\quad + (0.4) \left(95 \begin{bmatrix} -0.1 \\ -0.5 \\ 0.6 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right) \left(95 \begin{bmatrix} -0.1 \\ -0.5 \\ 0.6 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right)^T \\ &= \begin{bmatrix} 894.03 & -509.75 & -384.75 \\ -509.75 & 2352.75 & -1843 \\ -384.75 & -1843 & 2227.28 \end{bmatrix} \end{aligned}$$

(b.)

$$\text{Baseline } B(s) = 0.1 \times 100 + 0.5 \times 98 + 0.4 \times 95 = 97.$$

$$E[\hat{\nabla} V] = (0.1)(100-97) \begin{bmatrix} -0.1 \\ -0.5 \\ -0.4 \end{bmatrix} + (0.5)(98-97) \begin{bmatrix} -0.1 \\ 1-0.5 \\ -0.4 \end{bmatrix} + (0.4)(95-97) \begin{bmatrix} -0.1 \\ -0.5 \\ 1-0.4 \end{bmatrix} = \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix}$$

$$\begin{aligned} \text{Var}[\hat{\nabla} V] &= (0.1) \left(3 \begin{bmatrix} 0.9 \\ -0.5 \\ -0.4 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right) \left(3 \begin{bmatrix} 0.9 \\ -0.5 \\ -0.4 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right)^T \\ &\quad + (0.5) \left(1 \begin{bmatrix} -0.1 \\ 0.5 \\ 0.4 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right) \left(1 \begin{bmatrix} -0.1 \\ 0.5 \\ 0.4 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right)^T \\ &\quad + (0.4) \left(-2 \begin{bmatrix} -0.1 \\ -0.5 \\ 0.6 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right) \left(-2 \begin{bmatrix} -0.1 \\ -0.5 \\ 0.6 \end{bmatrix} - \begin{bmatrix} 0.3 \\ 0.5 \\ -0.8 \end{bmatrix} \right)^T \\ &= \begin{bmatrix} 0.66 & -0.5 & -0.16 \\ -0.5 & 0.5 & 0 \\ -0.16 & 0 & 0.16 \end{bmatrix} \end{aligned}$$

(c.)

Assume $B(s) = x$

trace of $\text{Var}[\hat{\nabla} V] =$

$$\Rightarrow (0.1) \left\{ [(100-b)(0.9) - 0.3]^2 + [(100-b)(-0.5) - 0.5]^2 + [(100-b)(-0.4) + 0.8]^2 \right\}$$

$$+ (0.5) \left\{ [(98-b)(-0.1) - 0.3]^2 + [(98-b)(0.5) - 0.5]^2 + [(98-b)(-0.4) + 0.8]^2 \right\}$$

$$+ (0.4) \left\{ [(95-b)(-0.1) - 0.3]^2 + [(95-b)(-0.5) - 0.5]^2 + [(95-b)(0.6) + 0.8]^2 \right\}$$

$f'(b) = 0$ when $b = 97.138$, so 97.138 is the optimal baseline.

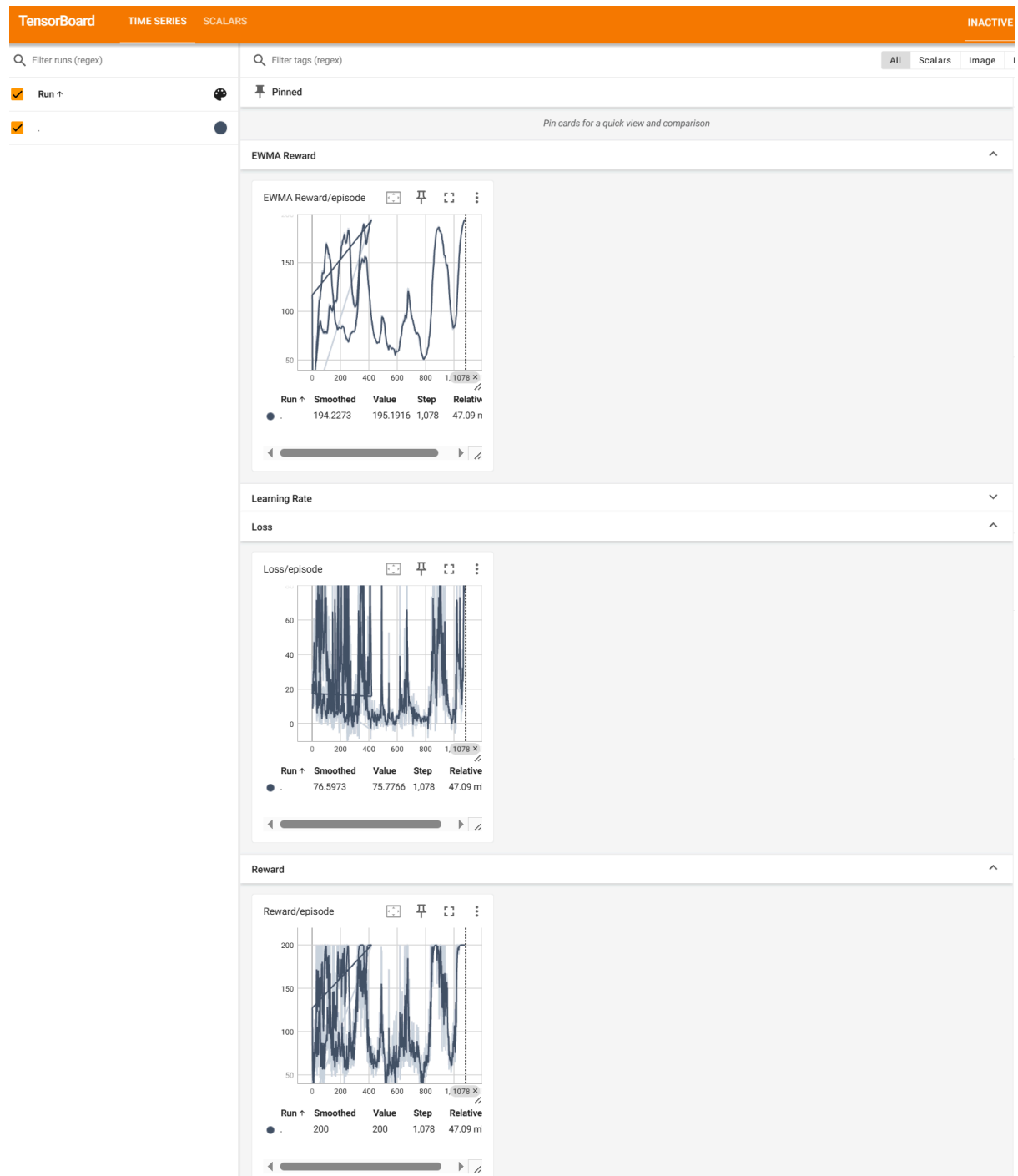
Problem4

(a)

NN architecture:

```
# Network layers
self.shared_fc = nn.Linear(self.observation_dim, self.hidden_size)
self.action_head = nn.Linear(self.hidden_size, self.action_dim)
self.value_head = nn.Linear(self.hidden_size, 1)
```

LR = 0.01



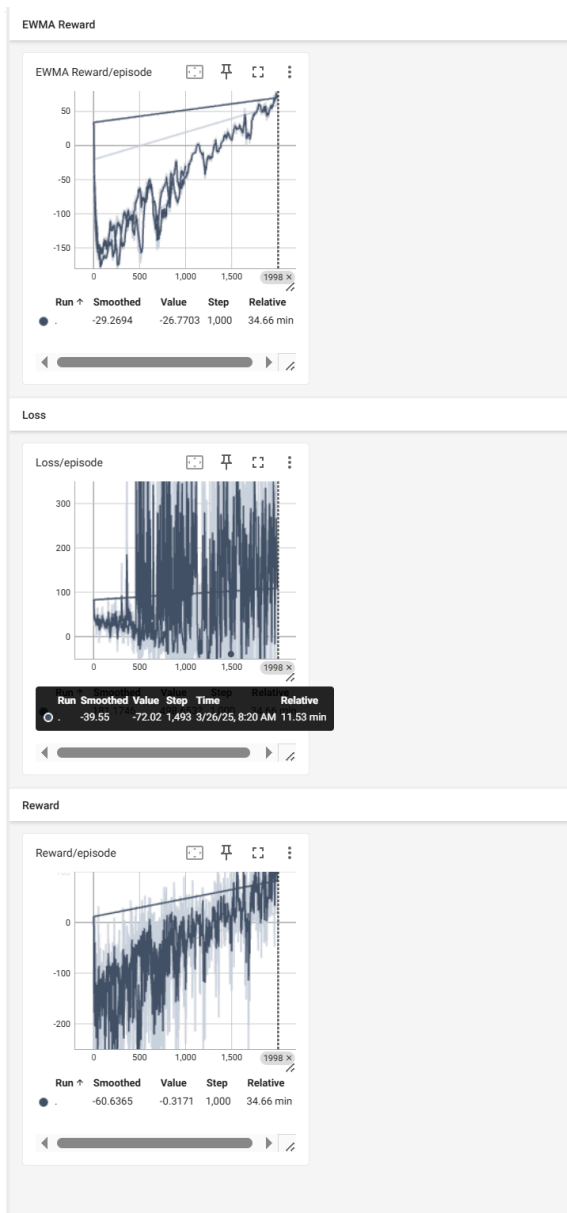
Training Summary

- **EWMA Reward:** Peaks at ~194, stabilizes near-optimal performance around episode 900.
- **Reward per Episode:** Frequent spikes to 200 (max), showing successful learning despite high variance.
- **Loss:** Spiky and unstable, typical for REINFORCE with Monte Carlo returns.

(b)

LR:1e-3

NN architecture: The same with (a)



Training Summary

- **EWMA Reward** steadily improves from around -150 to over +50, showing clear learning progress.
- **Loss** is highly variable, which is expected in policy gradient methods with GAE.
- **Raw Episode Rewards** trend upward, indicating that the agent is gradually mastering the LunarLander-v2 task.

(c)

GAE design:

parameters

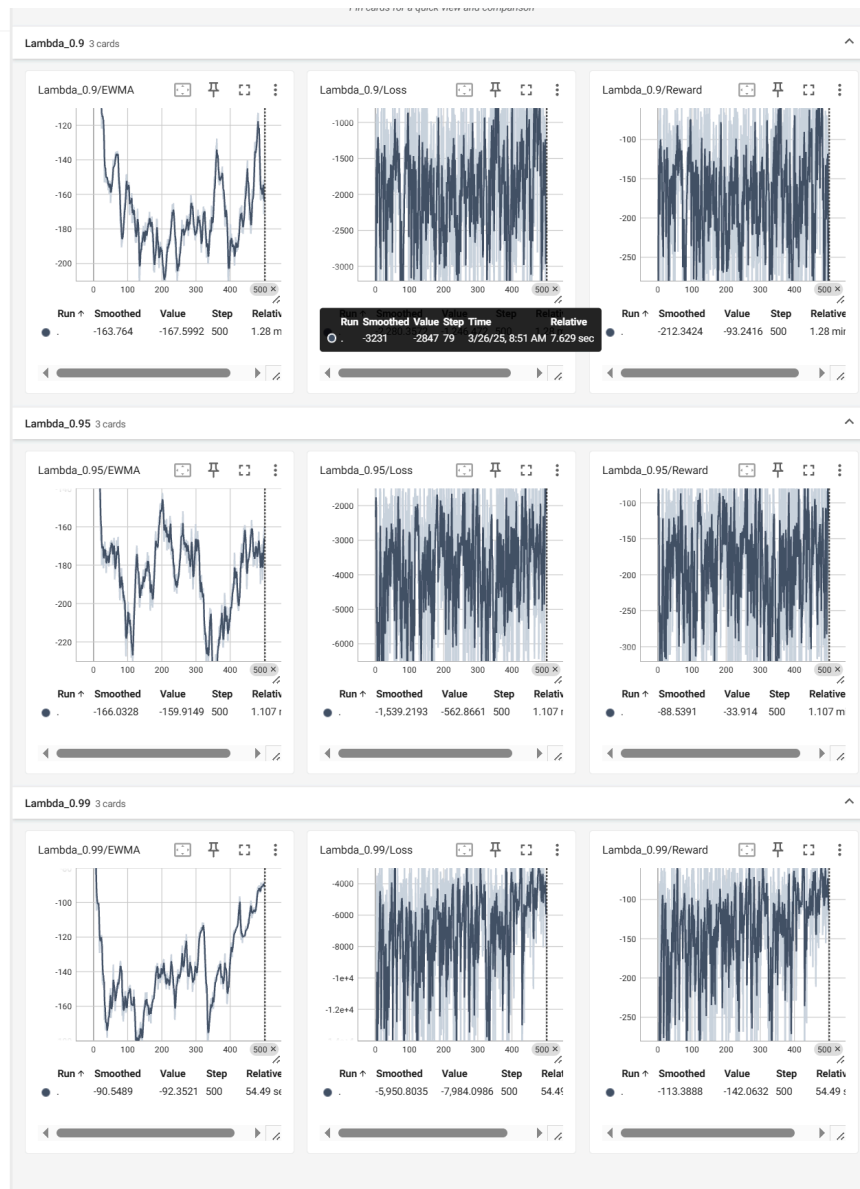
gamma = 0.99, lambda = 0.90/0.95/0.99

```
class GAE:
    def __init__(self, gamma=0.99, lambda_=0.95):
        self.gamma = gamma
        self.lambda_ = lambda_

    def compute_advantages(self, rewards, values, done):
        advantages = []
        gae = 0
        values = values + [0] # Add terminal value
        for t in reversed(range(len(rewards))):
            delta = rewards[t] + self.gamma * values[t + 1] * (1 - done) - values[t]
            gae = delta + self.gamma * self.lambda_ * (1 - done) * gae
            advantages.insert(0, gae)
        return advantages
```

compute_advantages

- **Delta:** We compute TD error as mentioned in the lecture.
- **Reversed time:** Compute GAE in reversed time.



$\lambda = 0.90$ shows unstable learning with no clear improvement in EWMA reward, and high variance in both reward and loss.

$\lambda = 0.95$ performs slightly better, but still lacks consistent upward reward trends—loss remains volatile.

$\lambda = 0.99$ demonstrates the most promising EWMA reward progression among the three, despite higher variance and larger magnitude losses.

Problem 5

Dataset Format

The dataset returned by D4RL is a dictionary with keys:

- **`observations`**: state vectors
- **`actions`**: corresponding actions
- **`rewards`**: scalar rewards
- **`terminals`**: flags indicating episode ends
- **`next\_observations`**: (via ``qlearning_dataset``) the subsequent state after each transition

We also generated a dataset from the ``hopper-medium-v2`` MuJoCo environment. The dataset format is the same but with 11D states and 3D continuous actions. This dataset simulates a medium-performance policy collected offline.